SIMATS ENGINEERING
ASSESSMENT TOOL 3

COMPUTER NETWORKS – COMPARATIVE ANALYSIS

Course Name: Computer Networks
Course Code: CSA0704
Course Outcome: CO4

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.

Introduction

Star and bus are two different network topologies. In star topology, every host has a separate connection to a central switch or hub. In bus topology, all devices share one common backbone cable.

Cost

Star topology normally costs more initially because it needs a central switch and a separate cable for each host. Bus topology uses a common backbone, so less cable and fewer connection points are required. However, the lower initial cost of bus topology may be offset by troubleshooting and maintenance difficulties.

Reliability

Star topology provides better reliability for individual link failures. If one host cable fails, normally only that host is disconnected while other hosts continue communicating. The main weakness is the central switch: if it fails, many or all connected hosts can lose communication.

Bus topology depends on a shared backbone. A fault in the backbone or its termination can affect many devices or the entire network. Therefore, bus topology has a greater impact when a shared component fails.

Performance

Star topology generally gives better performance because modern switches can forward frames to the required port instead of sending all traffic to every host. Each host also has a dedicated link to the switch.

Bus topology uses a shared medium. As more devices transmit, contention and traffic increase. This can reduce throughput and increase delays.

Parameter	Star	Bus
Initial cost	Moderate/High	Low
Cabling	Separate cable per host	Single backbone
Reliability	High for link failures	Lower
Performance	Generally high	Decreases with traffic
Troubleshooting	Easy to isolate	Backbone faults are harder
Expansion	Easy with switch ports	Limited by shared medium

Example

In a 20-computer laboratory using star topology, failure of one computer cable normally affects only that computer. With a bus arrangement, a fault in the shared backbone can interrupt communication for several or all computers.

Conclusion

Star topology is generally better for modern LANs because it provides better performance, reliability, fault isolation, and easier expansion. Bus topology is cheaper and simpler for small networks but is less suitable for large or high-traffic environments.

2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.

Introduction

Fault tolerance is the ability of a network to continue working when a device or communication link fails. Mesh and star topologies differ significantly in the amount of redundancy they provide.

Mesh Topology

In a full mesh, each node has a direct connection to every other node. This creates multiple possible paths between devices. If one link fails, data can use another path. Hence, mesh provides very high fault tolerance and redundancy.

The disadvantage is that the number of links grows rapidly. A full mesh with n devices requires $n(n-1)/2$ links. This increases cabling, port requirements, cost, installation time, and maintenance effort.

Star Topology

In star topology, every host connects to a central switch. If one access cable fails, normally only that host is affected. However, the central switch is a critical point. If it fails, all connected hosts may lose communication.

Feature	Mesh	Star
Fault tolerance	Very high	Moderate/High
Alternative paths	Multiple	Usually one
Single central failure	No central dependency	Switch can affect all hosts
Redundancy	Very high	Limited in basic design
Cost	Very high	Moderate
Maintenance	Complex	Relatively easy

Example

For critical servers, mesh connectivity can allow traffic to continue through another path when one link fails. In a basic star network, a switch failure can disconnect every attached server. This shows the higher inherent fault tolerance of mesh.

Conclusion

Mesh topology is more fault tolerant because it provides multiple paths and high redundancy. Star topology is more practical for normal LANs because it provides good reliability with much lower cost and complexity.

3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.

Introduction

Installation complexity depends on the amount of cabling, number of connections, equipment, configuration, and physical planning required.

Bus Topology

Bus topology is simple because all devices share a common backbone cable. It requires fewer physical connections and less cabling. It can therefore be installed quickly for small networks.
However, the backbone must be installed and terminated correctly. A backbone fault can affect many devices and may be difficult to locate.

Mesh Topology

Mesh topology is much more complex. In a full mesh, every node connects directly to every other node. The number of links is $n(n-1)/2$. For 5 devices, 10 links are required; for 10 devices, 45 links are required.
The rapid increase in links makes mesh expensive and time-consuming to install. It also requires careful cable management, ports, configuration, and troubleshooting.

Parameter	Bus	Mesh
Cabling	Low	Very high
Connections	Few	Many
Installation time	Low	High
Configuration	Simple	Complex
Cost	Low	High
Fault tolerance	Low	Very high
Maintenance	Simple structure	Complex

Conclusion

Bus topology is easier, faster, and cheaper to install. Mesh topology is much more complex but provides excellent redundancy and fault tolerance. Therefore, bus is preferred for simplicity, while mesh is selected when reliability is more important than installation cost.

4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.

Introduction
Scalability is the ability of a network to grow by adding users, devices, or network segments without requiring a complete redesign.

Star Topology
A star network can be expanded by adding hosts to unused switch ports. If ports are full, another switch can be connected. This makes star topology reasonably scalable for small and medium networks.
However, expansion depends on switch capacity, uplink bandwidth, available ports, cabling, and network design.

Hybrid Topology
Hybrid topology combines two or more topology types, such as star, bus, and mesh. Different departments or buildings can use different designs. New network segments can be added according to their requirements.
This flexibility makes hybrid topology suitable for universities, hospitals, enterprises, and large campuses where different areas may have different network needs.

Feature	Hybrid	Star
Scalability	Very high	High
Flexibility	Very high	Moderate
Expansion	Add network segments	Add hosts/switches
Complexity	High	Low/Moderate
Cost	Variable/High	Moderate
Best suited for	Large networks	Small/Medium LANs

Example
A university can use star topology inside individual departments and connect those departmental networks using a larger backbone structure. A new department can be added as another network segment without replacing the entire existing network.

Conclusion
Star topology provides good scalability for ordinary LAN expansion. Hybrid topology provides greater flexibility because different topology types can be combined and

expanded independently. Therefore, hybrid topology is more suitable for large and growing organizations.

5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

Introduction

Network maintenance includes monitoring devices, troubleshooting faults, replacing cables or equipment, checking connectivity, and maintaining reliable communication.

Star Topology

Star topology is relatively easy to maintain because each host has an identifiable connection to the central switch. A faulty cable or host can usually be isolated quickly. The main maintenance concern is the central switch, which can affect many devices if it fails.

Mesh Topology

Mesh topology is highly reliable but difficult to maintain because it contains many links and ports. Administrators may need to monitor numerous connections. The advantage is that a single link failure may not stop communication because another route can be available.

Bus Topology

Bus topology has a simple physical structure, but maintenance can be difficult when the backbone develops a fault. Because many devices share the same cable, locating the exact fault can take time and a backbone failure can affect many hosts.

Hybrid Topology

Hybrid topology has moderate to high maintenance complexity depending on the topologies combined. Administrators must understand different network segments and devices. However, its modular design can help isolate a problem to one particular segment.

Topology	Maintenance	Main Issue	Main Advantage
Star	Easy	Central switch dependency	Easy fault isolation
Mesh	Difficult	Many links and ports	High redundancy
Bus	Moderate/Difficult	Backbone failure	Simple structure
Hybrid	Moderate/High	Multiple topology types	Flexible and modular

Example

If one cable fails in a star network, the affected host can normally be identified quickly. In a bus network, a backbone problem may affect several devices. In a mesh network, communication may continue through another path, but the failed link still needs to be located and repaired. In a hybrid network, administrators first identify the affected segment and then troubleshoot that segment.

Conclusion

Star topology provides the best balance of reliability and maintenance simplicity for most LANs. Mesh provides excellent fault tolerance but requires more maintenance. Bus has a simple structure but backbone faults can be difficult to troubleshoot. Hybrid topology provides flexibility but requires administrators to manage multiple topology types.

Overall Comparative Summary

Criteria	Star	Bus	Mesh	Hybrid
Cost	Moderate	Low	Very High	Variable/High
Reliability	High	Low/Moderate	Very High	High
Performance	High	Low/Moderate	High	High
Fault tolerance	Moderate/High	Low	Very High	Depends on design
Installation	Easy/Moderate	Easy	Difficult	Complex
Scalability	High	Low	Limited by link growth	Very High
Maintenance	Easy	Moderate/Difficult	Difficult	Moderate/High

Final Conclusion

Star, bus, mesh, and hybrid topologies have different advantages and limitations. Star is generally the most practical choice for modern LANs because it combines good performance, reliability, easy troubleshooting, and reasonable scalability. Bus is inexpensive and simple but has limitations because all devices share one backbone. Mesh provides the highest redundancy and fault tolerance but requires more cables, cost, and maintenance. Hybrid topology offers the greatest flexibility and scalability by combining different topologies according to organizational requirements.

Therefore, topology selection should consider network size, budget, traffic, reliability requirements, future expansion, installation complexity, and maintenance capability. For a typical college or office LAN, star topology is usually a practical choice, while mesh or hybrid designs are useful when high availability, redundancy, or large-scale expansion is required.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Understanding	15%	Fully understands the problem and requirements; no clarification needed	Understands most requirements with minor clarification	Partial understanding; misses some requirements	Poor understanding of the problem
Algorithm Design	20%	Efficient, optimal, and well-structured algorithm	Correct algorithm with minor inefficiencies	Basic approach but not optimal	Incorrect or no clear algorithm
Code Quality	15%	Clean, well-organized, readable, and properly formatted code	Mostly clean code with minor issues	Code is somewhat unclear or inconsistent	Poorly written, unreadable code
Functionality	20%	Code works perfectly for all test cases	Works for most test cases	Works for limited cases	Does not run or fails major cases
Error Handling	10%	Handles all edge cases and errors effectively	Handles some edge cases	Limited error handling	No error handling
Efficiency	10%	Highly optimized in time and space complexity	Reasonably efficient	Some inefficiencies	Highly inefficient
Documentation & Comments	5%	Clear and meaningful comments and documentation	Adequate comments	Few or unclear comments	No comments
Testing & Debugging	5%	Thoroughly tested with multiple cases; no bugs	Tested with some cases	Minimal testing	No testing done